

Question			Marking details	Marks available																																
				AO1	AO2	AO3	Total	Maths	Prac																											
7	(a)		Indicative content: <table><tr><th rowspan="2">Section</th><th colspan="3">Descriptions</th></tr><tr><th>Qualitative</th><th>Numerical</th><th>Developed</th></tr><tr><td>AB</td><td>Acceleration</td><td>0 to 25 m/s in 50 s</td><td>Constant acceleration (of 0.5 m/s²)</td></tr><tr><td>BC</td><td>Constant velocity</td><td>25 m/s for 80 s</td><td></td></tr><tr><td>CD</td><td>Deceleration</td><td>25 m/s to 0 in 70 s</td><td>Increasing deceleration</td></tr><tr><td>DE</td><td>Stationary</td><td>For 40 s</td><td></td></tr><tr><td>EF</td><td>Acceleration</td><td>0 to 10 m/s in 60 s [or from 240 s – 300 s]</td><td>Constant acceleration less than AB (0.17 m/s²)</td></tr></table> 5 – 6 marks Correctly describes the whole motion with a good coverage of all three description areas <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3 – 4 marks Gives a detailed description of the motion which involves two of the three description areas or a partial content of all three areas. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i>	Section	Descriptions			Qualitative	Numerical	Developed	AB	Acceleration	0 to 25 m/s in 50 s	Constant acceleration (of 0.5 m/s ²)	BC	Constant velocity	25 m/s for 80 s		CD	Deceleration	25 m/s to 0 in 70 s	Increasing deceleration	DE	Stationary	For 40 s		EF	Acceleration	0 to 10 m/s in 60 s [or from 240 s – 300 s]	Constant acceleration less than AB (0.17 m/s ²)	3	3		6	3	
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				1-2 marks Some relevant points made from any of the description areas. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks No attempt made or no response worthy of credit.						
	(b)	(i)		Substitution: Momentum change = $10\,000 \times (25 - 0)(1)$ = $250\,000$ [kg m/s] (1)	1	1		2	2	
		(ii)		Substitution: $F = \frac{\Delta p}{t} = \frac{250\,000(\text{ecf})}{50}(1)$ = $5\,000$ [N](1) Alternative: $a = 25 \div 50 = 0.5$ [m/s ²] (1) $F = ma = 0.5 \text{ ecf} \times 10\,000 = 5\,000$ [N] (1) [NB. ecf only on a <u>calculation</u> of acceleration, not on a change in velocity]	1	1		2	2	
		(iii)		Substitution: Distance = 25×80 (1) [No tolerance] = $2\,000$ [m] (1)	1	1		2	2	
		(iv)		The line / it would be lower on the graph [than BC] (1) The length [of the line] would be greater [than 80 s](1)			2	2	1	
	(c)			Bigger mass / more inertia [not: heavier] (1) [so the same force] would produce a <u>smaller</u> [not: slower] acceleration (1)	2			2		
				Question 7 total	8	6	2	16	10	0